Wagner S Chittó - Homework 8

Calculos e simulações do artigo "Simplified Analysisi of Feedback Amplifiers"

Amp. Series-shunt (Fig.5)

$\beta = \frac{V_f}{V_o} = \frac{1}{11}K\Omega$

$A_F = \frac{1}{\beta} = 11$ ✓

$\beta = 100$

$r_{e_1} + r_{e_2} = 50\Omega$

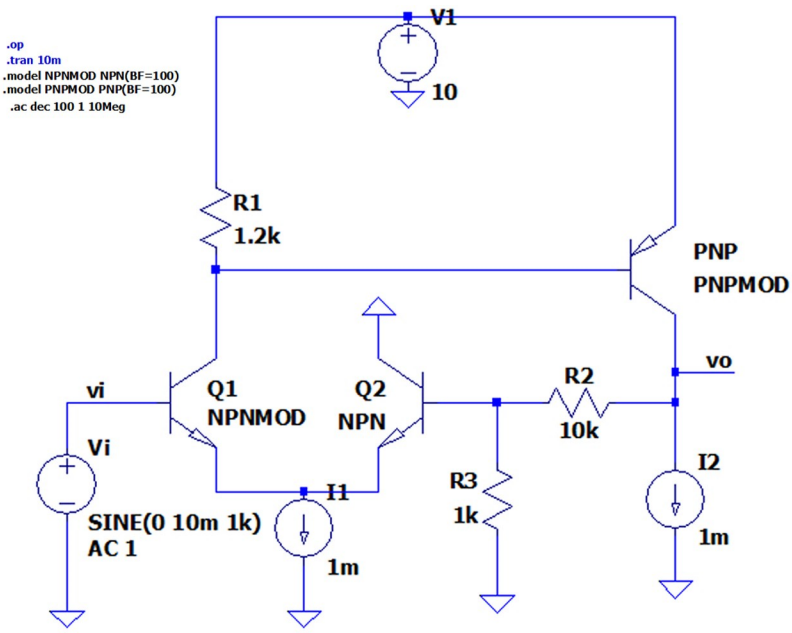
$R = (\beta(r_{e_1}+r_{e_2}) \| 1k + 10k = 10{,}9k\Omega$

$T = 295 >> 1$ ✓

$R_i = 10{,}9K$

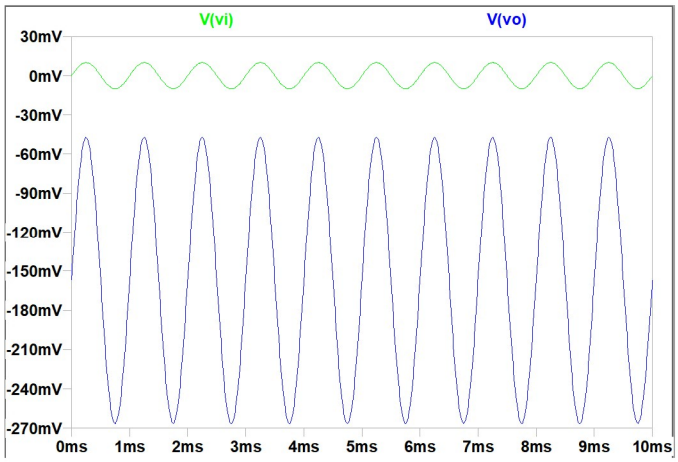
$R_{if} = R_i(1+T) = 3M\Omega$

$R_{of} = \frac{R_o}{1+T} = 37\Omega$ ✓



--- Operating Point ---

V(vi):	0	voltage
V(n003):	-0.0173058	voltage
V(vo):	-0.156829	voltage
V(n002):	9.22613	voltage
V(n004):	-0.763277	voltage
V(n001):	10	voltage
Ic(Q1):	0.00065475	device_current
Ib(Q1):	6.5475e-06	device_current
Ie(Q1):	-0.000661298	device_current
Is(Q1):	0	device_current
I(I1):	0.001	device_current
Ic(PNPMOD):	-0.000986048	device_current
Ib(PNPMOD):	-9.86047e-06	device_current
Ie(PNPMOD):	0.000995908	device_current
Is(PNPMOD):	0	device_current
I(R3):	1.73058e-05	device_current
I(R1):	0.00064489	device_current
Ic(Q2):	0.000335349	device_current
Ib(Q2):	3.35349e-06	device_current
Ie(Q2):	-0.000338702	device_current
Is(Q2):	0	device_current
I(Vi):	-6.5475e-06	device_current
I(R2):	-1.39523e-05	device_current
I(I2):	0.001	device_current
I(V1):	-0.0016408	device_current



Amp. Series-series (Fig.7.7)

$R_1 = 9K\Omega$ $\quad R_3 = 600\Omega$

$R_2 = 5K\Omega$ $\quad R_{E1} = R_{E2} = 640\Omega$

$I_{C_1} = 0,6mA$ $\quad I_{C_3} = 4mA$

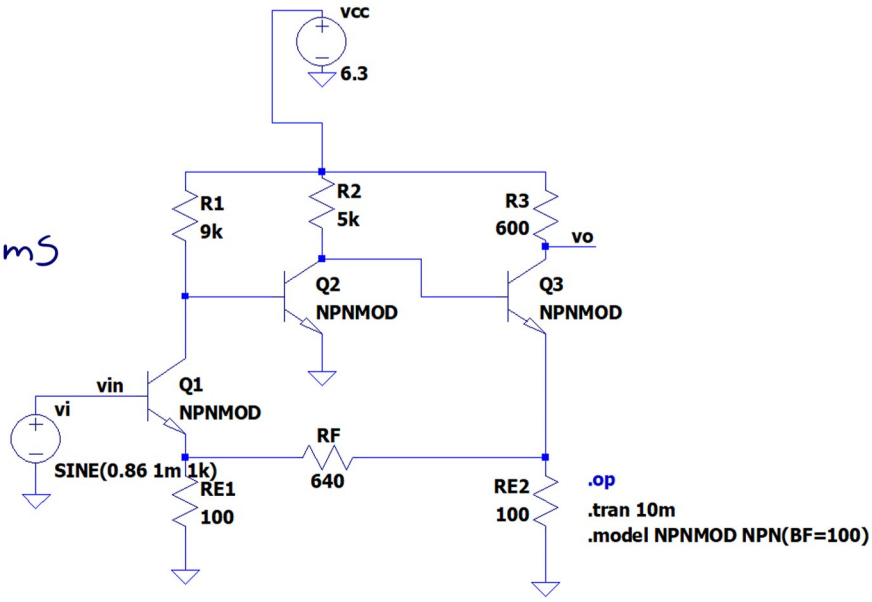
$I_{C_2} = 1mA$

$$\beta = \frac{R_{E1}R_{E2}}{R_{E1}+R_{E2}+R_F} \quad \beta = 11,7\Omega \quad \frac{v_o}{v_i} = \frac{1}{\beta} = 84mS$$

$$\frac{v_o}{v_i} = -\frac{R_3}{\beta} \quad A_v = -60,4$$

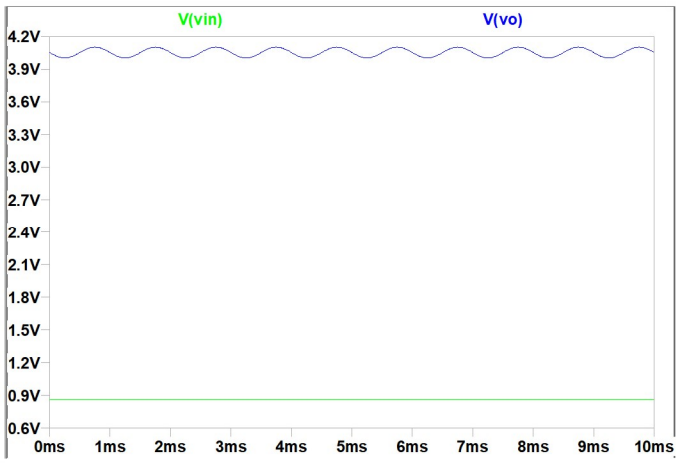
$T = 247 >> 1$ ✓

$R_1 = 13K\Omega$ $\quad R_{1f} = 3,2M\Omega$




```
        --- Operating Point ---

V(vin):         0.86            voltage
V(n002):        1.1491          voltage
V(vo):          4.05153         voltage
V(n001):        6.3             voltage
V(n003):        0.774042        voltage
V(n004):        0.0988064       voltage
V(n005):        0.340697        voltage
Ic(Q1):         0.000604068     device_current
Ib(Q1):         6.04068e-06     device_current
Ie(Q1):         -0.000610109    device_current
Is(Q1):         0               device_current
I(R2):          0.00103018      device_current
I(R3):          -0.00374746     device_current
I(vcc):         -0.00539163     device_current
Ic(Q3):         0.00374746      device_current
Ib(Q3):         3.74746e-05     device_current
Ie(Q3):         -0.00378493     device_current
Is(Q3):         0               device_current
I(R1):          0.000613995     device_current
Ic(Q2):         0.000992706     device_current
Ib(Q2):         9.92706e-06     device_current
Ie(Q2):         -0.00100263     device_current
Is(Q2):         0               device_current
I(RE1):         0.000988064     device_current
I(RF):          0.000377955     device_current
I(RE2):         -0.00340697     device_current
I(vi):          -6.04068e-06    device_current
```



Amp. Shunt-shunt (Fig.15)

$\frac{V_o}{V_i} = \frac{R_F + R_{E1}}{R_S} = A_v = 7,4$

Resto igual ao anterior.

```
.op
.tran 10m
.model NPNMOD NPN(BF=100)
```

